Lab 4 Report: Installing Gitea via Docker and Demonstrating LFS Support

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Objective

Install Gitea using Docker on a local machine.
Verify that Gitea supports Git LFS (Large File Storage).
Create a repository containing a file larger than 1 GB and push it using Git LFS.

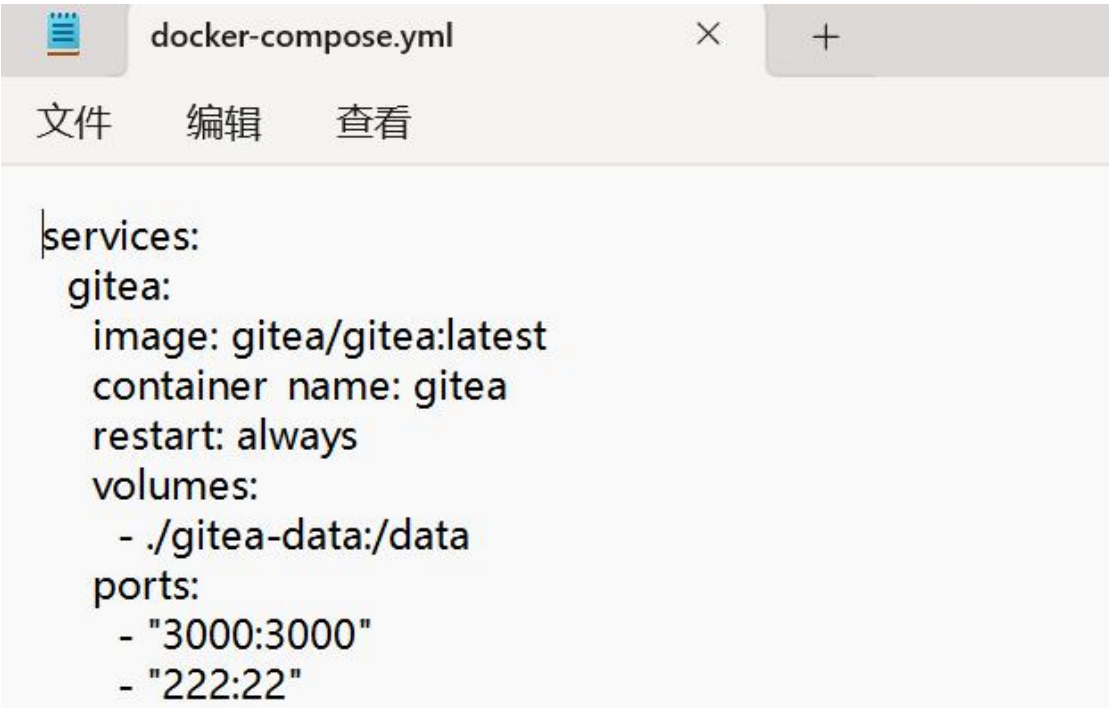
Environment

OS: Windows 11 with WSL2 and Docker Desktop
Tools: Docker Compose, Git for Windows, Git LFS

Part A: Install Gitea Using Docker

Step 1: Prepare Docker Compose file

Created docker-compose.yml in E:\云计算导论\Lab1:



Step 2: Start Gitea container

bash

```
docker-compose up -d
```

Output shows that the image was pulled and the container started successfully.

```
PS E:\云计算导论\Lab1> docker-compose up -d
[+] up 9/9
 ✔Image gitea/gitea:latest Pulled                                    48.5s
 ✔Network lab1_default     Created                                    0.0s
 ✔Container gitea          Started                                    0.5s
PS E:\云计算导论\Lab1>
```

Step 3: Initial Gitea configuration

Accessed `http://localhost:3000` and completed the installation:

Database: SQLite
Site title: My Gitea
Base URL: http://localhost:3000
Admin account: lwc1116 / [password]

Part B: Verify LFS Support

Step 1: Enable LFS in Gitea configuration

Entered the Gitea container and modified `app.ini`:

bash

```bash
docker exec -it gitea /bin/bashvi /data/gitea/conf/app.ini
```

Added under `[server]`:

ini

```ini
LFS_START_SERVER = true
```

Saved and restarted the container:

bash

```bash
docker restart gitea
```

```
PS E:\云计算导论\Lab1> docker exec -it gitea /bin/bash
f22879f67d58:/# vi /data/gitea/conf/app.ini
f22879f67d58:/# exit
exit
PS E:\云计算导论\Lab1> docker restart gitea
gitea
PS E:\云计算导论\Lab1> docker exec -it gitea cat /data/gitea/conf/app.ini | grep LFS_START_SERVER
grep : 无法将“grep”项识别为 cmdlet、函数、脚本文件或可运行程序的名称。请检查名称的拼写，如果包括路径，请确保路径正确
，然后再试一次。
所在位置 行:1 字符: 54
+ docker exec -it gitea cat /data/gitea/conf/app.ini | grep LFS_START_S ...
+                                                      ~~~~
    + CategoryInfo          : ObjectNotFound: (grep:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS E:\云计算导论\Lab1> docker exec -it gitea cat /data/gitea/conf/app.ini | findstr LFS_START_SERVER
LFS_START_SERVER = true
LFS_START_SERVER = true
PS E:\云计算导论\Lab1>
```

Step 2: Install Git LFS client locally

bash

```bash
git lfs install
```

```
Windows PowerShell
Windows PowerShell
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安装最新的 PowerShell，了解新功能和改进！ https://aka.ms/PSWindows

PS E:\云计算导论\Lab1> git --version
git version 2.54.0.windows.1
PS E:\云计算导论\Lab1> git lfs version
git-lfs/3.7.1 (GitHub; windows amd64; go 1.25.1; git b84b3384)
PS E:\云计算导论\Lab1>
```

Step 3: Create a test repository and push a 10 MB file using LFS

Created repository `lfs-test` on Gitea web.
Local commands:

bash

cd E:\gitea-testgit initgit lfs track "*.bin"git add .gitattributesgit commit -m "Track .bin files with LFS"fsutil file createnew test.bin 10485760git add test.bingit commit -m "Add 10MB test file via LFS"git remote add origin http://localhost:3000/lwc1116/lfs-test.gitgit push -u origin main

Successful push output included:

text

Uploading LFS objects: 100% (1/1), 10 MB | 0 B/s, done.

```
PS C:\Users\21315> cd E:\
PS E:\> mkdir gitea-test


    目录: E:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2026/4/26     15:56                gitea-test


PS E:\> cd gitea-test
PS E:\gitea-test> git init
Initialized empty Git repository in E:/gitea-test/.git/
PS E:\gitea-test> git lfs track "*.bin"
Tracking "*.bin"
PS E:\gitea-test> git add .gitattributes
PS E:\gitea-test> git commit -m "Track .bin files with LFS"
Author identity unknown

*** Please tell me who you are.

Run
```

```
  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got '21315@◆◆◆◆◆◆.(none)')
PS E:\gitea-test> git config --global user.name "lwc1116"
PS E:\gitea-test> git config --global user.email "2131545006@qq.com"
PS E:\gitea-test> git commit -m "Track .bin files with LFS"
[main (root-commit) 3870dbf] Track .bin files with LFS
 1 file changed, 1 insertion(+)
 create mode 100644 .gitattributes
PS E:\gitea-test> fsutil file createnew test.bin 10485760
已创建文件 E:\gitea-test\test.bin
PS E:\gitea-test> git add test.bin
PS E:\gitea-test> git commit -m "Add 10MB test file via LFS"
[main 8b5a7de] Add 10MB test file via LFS
 1 file changed, 3 insertions(+)
 create mode 100644 test.bin
PS E:\gitea-test> git remote add origin http://localhost:3000/lwc1116/lfs-test.git
```

```
PS E:\gitea-test> git push -u origin main
warning: use of unencrypted HTTP remote URLs is not recommended; see https://aka.ms/gcm/unsaferemotes for more informati
on.
Locking support detected on remote "origin". Consider enabling it with:
  $ git config lfs.http://localhost:3000/lwc1116/lfs-test.git/info/lfs.locksverify true
Uploading LFS objects: 100% (1/1), 10 MB | 0 B/s, done.
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 24 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 630 bytes | 630.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/lwc1116/lfs-test.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS E:\gitea-test>
```

Part C: Create a Repository with a 1 GB+ File

Step 1: Generate a 1 GB file

bash

```
cd E:\gitea-largefsutil file createnew large.bin 1073741824
```

```
PS E:\gitea-test> cd E:\
PS E:\> mkdir gitea-large


    目录: E:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         2026/4/26     16:06                gitea-large


PS E:\> cd gitea-large
PS E:\gitea-large>
PS E:\gitea-large> fsutil file createnew large.bin 1073741824
已创建文件 E:\gitea-large\large.bin
```

Step 2: Create a new repository on Gitea named large-repo

Step 3: Clone, configure LFS, and push the 1 GB file

bash

```
git clone http://localhost:3000/lwc1116/large-repo.gitcd large-repogit lfs track "*.bin"git add .gitattributesgit commit -m "Track .bin files"copy ..\large.bin .git add large.bingit commit -m "Add 1GB file via LFS"git push -u origin main
```

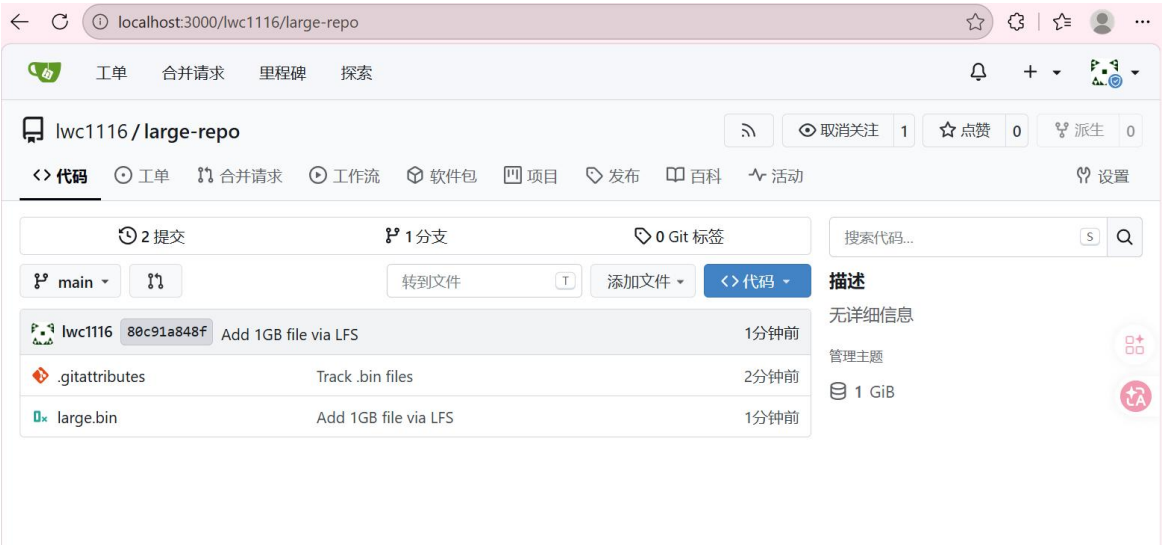
Push output:

text

Uploading LFS objects: 100% (1/1), 1.1 GB | 124 MB/s, done.* [new branch] main -> main

```
PS E:\gitea-large> git clone http://localhost:3000/lwc1116/large-repo.git
Cloning into 'large-repo'...
remote: Repository not found
fatal: repository 'http://localhost:3000/lwc1116/large-repo.git/' not found
PS E:\gitea-large> git clone http://localhost:3000/lwc1116/large-repo.git
Cloning into 'large-repo'...
warning: You appear to have cloned an empty repository.
```

```
PS E:\gitea-large> cd E:\gitea-large\large-repo
PS E:\gitea-large\large-repo> git lfs track "*.bin"
Tracking "*.bin"
PS E:\gitea-large\large-repo> git add .gitattributes
PS E:\gitea-large\large-repo> git commit -m "Track .bin files"
[main (root-commit) 0d18983] Track .bin files
 1 file changed, 1 insertion(+)
 create mode 100644 .gitattributes
PS E:\gitea-large\large-repo> copy ..\large.bin .
PS E:\gitea-large\large-repo> git add large.bin
PS E:\gitea-large\large-repo> git commit -m "Add 1GB file via LFS"
[main 80c91a8] Add 1GB file via LFS
 1 file changed, 3 insertions(+)
 create mode 100644 large.bin
PS E:\gitea-large\large-repo> git push -u origin main
Locking support detected on remote "origin". Consider enabling it with:
  $ git config lfs.http://localhost:3000/lwc1116/large-repo.git/info/lfs.locksverify true
Uploading LFS objects: 100% (1/1), 1.1 GB | 124 MB/s, done.
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 24 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 620 bytes | 620.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/lwc1116/large-repo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS E:\gitea-large\large-repo>
```



Conclusion

Gitea was successfully installed and run via Docker on a local Windows machine.

LFS functionality was enabled and verified by pushing a 10 MB file that was correctly tracked and uploaded as an LFS object.
A repository containing a 1 GB file was created and pushed using LFS, confirming that Gitea can handle large files efficiently.
All required objectives (a, b, c) have been fulfilled. The optional tasks (backup/restore and non- Docker installation) are not covered in this report.